

Bayesian Machine Learning in Quantitative Finance

Chapter 9: Detecting Unauthorized Expenditure

Based on Mongwe, Mbuva & Marwala

Langevin and Hamiltonian Monte Carlo for Public-Finance Oversight

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9.1 Introduction: Why Municipal Overspending Matters I

Intuition first. A municipality has a legally approved budget. If it spends money that was never authorized in that budget – or spends a conditional grant on something other than what it was earmarked for – that spending is a compliance failure, independent of whether anyone stole anything. Detecting this *before* the annual audit is a genuine public-finance oversight problem, and it is exactly the kind of structured-data classification problem where Bayesian machine learning can help.

The South African context

The Auditor-General of South Africa (AGSA) found that **68% of municipalities** expended **R25.47 billion** in unauthorized spending in the 2021–2022 fiscal year. South Africa has 257 municipalities (8 metropolitan, 44 district, 205 local), constitutionally mandated to deliver services such as water, sanitation, power, roads, and refuse collection – funded partly through own revenue and partly through conditional/unconditional grants from provincial and national government.

9.1 Introduction: Why Municipal Overspending Matters II

What this chapter does

- Builds a **Bayesian logistic regression** model that predicts, from a municipality's *unaudited* financial ratios, whether its *audited* statements will later be flagged for unauthorized expenditure.
- Trains this model with four MCMC samplers: MH, MALA, HMC, and **Magnetic HMC (MHMC)** – the new method introduced in this chapter.
- Uses an automatic relevance determination (ARD) prior so the model also *ranks which financial ratios matter most* – useful for auditors deciding where to focus limited audit resources.

Why this matters for risk-based auditing: previous work (Sikhosana, 2021) only established statistical *correlations* between unlawful spending and financial ratios computed from the *same* audited statements as the target – i.e., it could not be used before the audit was performed. This chapter is presented as the first-in-literature attempt to *predict* unauthorized expenditure using only unaudited (pre-audit) financials, turning this into a genuine risk-screening tool.

9.2 Background to Unlawful Expenditure Analysis I

Intuition. South Africa's Public Finance Management Act (PFMA, 1999) requires government entities to spend efficiently and within budget. When the Auditor-General reviews a municipality's books, any spending that breaks these rules is labeled *unlawful*, and it comes in four flavors.

Four categories of unlawful expenditure

- **Unauthorized expenditure:** money spent that was not budgeted for, or that did not match the terms of a grant. (*The focus of this chapter.*)
- **Irregular expenditure:** spending not incurred in the manner required by law – does *not* always imply money was wasted or fraud occurred, just that the correct procurement/process rules were not followed.
- **Fruitless and wasteful expenditure:** spending made in vain that reasonable care could have avoided.
- **Irregular, wasteful, fruitless & unauthorized** are collectively termed **unlawful expenditure**.

9.2 Background to Unlawful Expenditure Analysis II

Why financial ratios should help

Sikhosana (2021) found statistically significant relationships between unlawful spending and financial-performance ratios: e.g., a *negative* link between unauthorized expenditure and current ratio, cost coverage, and net operating surplus margin. This tells us the ratios carry signal – but that analysis only described correlations *after the fact*, using ratios computed from the audited statements themselves (the same source as the label). That is not useful as an early-warning tool.

This chapter's contribution

Use a **Bayesian** approach for two reasons: (1) it gives probabilistically sound predictions with **uncertainty bounds**, not just point classifications; (2) an ARD-style prior gives a natural **importance ranking** of the input ratios, which is directly interpretable to auditors deciding where to allocate limited audit resources.

Notation You'll Need / Efficient Recap I

You have already seen MH, MALA and HMC in earlier chapters of this series (Ch. 3, Ch. 7). Here is the compressed recap before we introduce what's **new** in this chapter: Magnetic HMC.

Metropolis–Hastings (MH) – one line

Propose $\mathbf{w}^* \sim \mathcal{N}(\mathbf{w}, \sigma^2 I)$ (a symmetric random walk around the current state), accept with probability $\min(1, \frac{p(\mathbf{w}^*|\mathcal{D})}{p(\mathbf{w}|\mathcal{D})})$. No gradient information is used, so proposals are unaware of which direction improves the posterior – this produces highly autocorrelated chains that need very large sample sizes.

MALA – Langevin dynamics + MH correction

$$d\mathbf{w}_t = \frac{1}{2}f'(\mathbf{w})dt + dZ_t \quad (9.3)$$

where $f(\mathbf{w})$ is the target log-posterior, $f'(\mathbf{w})$ is its first derivative (the drift, pulling proposals *uphill*), and Z_t is Brownian motion (the diffusive/exploratory part). Discretizing with Euler–Maruyama introduces error, so a Metropolis accept/reject step is added on top to guarantee the correct stationary distribution. MALA mixes faster than MH because the drift term exploits gradient information, but samples are still noticeably correlated.

HMC – Hamiltonian dynamics

Introduce an auxiliary momentum $\mathbf{p}$ and simulate a physical system with Hamiltonian $H(\mathbf{w}, \mathbf{p}) = -\log p(\mathbf{w}|\mathbf{D}) + \frac{1}{2}\mathbf{p}^\top M^{-1}\mathbf{p}$ (potential energy = negative log-posterior, kinetic energy = Gaussian in $\mathbf{p}$):

$$\frac{d\mathbf{w}}{dt} = \frac{\partial H(\mathbf{w}, \mathbf{p})}{\partial \mathbf{p}}, \quad \frac{d\mathbf{p}}{dt} = -\frac{\partial H(\mathbf{w}, \mathbf{p})}{\partial \mathbf{w}} \quad (9.4)$$

Integrated numerically with the **leapfrog** scheme, then corrected with a Metropolis accept step to cancel discretization error. Because HMC proposes distant, high-acceptance-probability points by following the energy-conserving trajectory, its samples are far less autocorrelated than MH or MALA. When the trajectory length is set to 1 leapfrog step, HMC reduces exactly to MALA.

Notation You'll Need / Efficient Recap IV

What's new: Magnetic Hamiltonian Monte Carlo (MHMC) – plain-language intuition

Ordinary HMC is a **marble rolling under gravity/potential forces alone**: at every point it only feels the pull of the landscape (the gradient of the negative log-posterior), so it swings back and forth along smooth, in-plane arcs.

Magnetic HMC adds an extra force that always pushes sideways – perpendicular to the current direction of motion – exactly like the Lorentz force on a charged particle moving through a magnetic field. A sideways force cannot speed a particle up or slow it down (it does no work), but it *bends* the path. This breaks the usual symmetric, purely-gradient-driven HMC trajectory in a controlled, still-valid way, and for some posteriors (especially ones with correlated or curved geometry) the resulting curving/spiraling exploration mixes faster than straight HMC sweeps.

9.3 Model for Bayesian Detection of Expenditure I

Intuition. We want a probabilistic yes/no classifier: given a municipality's unaudited financial ratios, what is the probability that its statements will show unauthorized expenditure? A **Bayesian logistic regression** is the natural choice: logistic regression handles the binary classification, and the Bayesian treatment gives us calibrated uncertainty plus automatic feature-importance ranking.

Negative log-likelihood (Eq. 9.1)

$$l(D|\mathbf{w}) = \sum_i^N y_i \log(\mathbf{w}^\top x_i) + (1 - y_i) \log(1 - \mathbf{w}^\top x_i)$$

where $D = \{\mathbf{x}, \mathbf{y}\}$ is the data, N is the number of observations, $\mathbf{w}$ are the parameters (regression weights) to be estimated, $y_i \in \{0, 1\}$ is the unauthorized-expenditure label for observation i , and x_i is its feature vector of financial ratios. (In practice $\mathbf{w}^\top x_i$ is passed through a sigmoid link so it lies in $(0, 1)$ before being interpreted as a probability.)

9.3 Model for Bayesian Detection of Expenditure II

Un-normalized log-posterior (Eq. 9.2)

$$\ln p(\mathbf{w}|\mathbf{D}) = l(\mathbf{D}|\mathbf{w}) + \ln p(\mathbf{w}|\alpha) + \ln q(\alpha)$$

where $\ln p(\mathbf{w}|\alpha)$ is the log-prior on the weights given hyperparameters α , and $\ln q(\alpha)$ is the distribution placed on those hyperparameters. This is the un-normalized target that every MCMC sampler in this chapter (MH, MALA, HMC, MHMC) draws samples from.

Automatic Relevance Determination (ARD) prior

Each weight w_i has **its own** Gaussian prior, zero mean, with its own standard deviation/regularization parameter α_i . Large $\alpha_i \Rightarrow$ feature i is important (its weight is free to move far from zero); $\alpha_i \rightarrow 0 \Rightarrow$ feature i is irrelevant (its weight is shrunk to zero). This automatically ranks feature importance as a by-product of inference – no separate feature-selection step needed. The α_i are themselves given a log-normal distribution (mean 0, unit variance), and $\mathbf{w}$ and α are sampled *jointly* rather than via a Gibbs step – this doubles the effective parameter space but gives more stable MCMC exploration in practice.

9.3 Model for Bayesian Detection of Expenditure III

Why joint sampling instead of Gibbs sampling for α ?

An alternative (Neal, 2012) alternates HMC for the weights $\mathbf{w}$ with Gibbs sampling of α from a Gamma posterior. This converges to the asymptotically correct joint posterior, but still requires design choices (e.g. the exact conjugate prior family). Jointly sampling $\mathbf{w}$ and α with the log-normal parameterization avoids this extra design step and, per the authors, gives more stable exploration – at the cost of a doubled parameter dimension for the sampler to explore.

Python: Bayesian Logistic Regression with an ARD Prior

```
1 import numpy as np
2
3 def sigmoid(z):
4     return 1.0 / (1.0 + np.exp(-z))
5
6 def neg_log_likelihood(w, X, y):
7     # Eq. 9.1:  $l(D/w)$  -- numerically stable form via log1p/logaddexp
8     z = X @ w
9     return -np.sum(y * (-np.logaddexp(0, -z))
10                    + (1 - y) * (-np.logaddexp(0, z)))
11
12 def log_prior(w, alpha):
13     # zero-mean Gaussian prior, one std dev alpha_i PER weight (ARD)
14     return -0.5 * np.sum((w ** 2) / (alpha ** 2)) - np.sum(np.log(alpha))
15
16 def log_hyperprior(alpha):
17     # alpha_i ~ LogNormal(mean=0, var=1)
18     return -0.5 * np.sum(np.log(alpha) ** 2) - np.sum(np.log(alpha))
19
20 def log_posterior(w, alpha, X, y):
21     # Eq. 9.2: joint (unnormalized) target for MCMC over (w, alpha)
22     return (-neg_log_likelihood(w, X, y)
23            + log_prior(w, alpha) + log_hyperprior(alpha))
```

9.4.1 Metropolis-Hastings and the Metropolis-Adjusted Langevin Algorithm I

Intuition. We now train the ARD Bayesian logistic regression model above with four different samplers. MH and MALA are the efficient recap (seen in previous chapters); the emphasis here is on how the book applies them to *this* expenditure-detection posterior.

MH: the baseline

Random-walk Metropolis-Hastings uses a Gaussian proposal centered at the current state. It ignores gradient information entirely, so its exploration is undirected – in this chapter's results, MH indeed produces the most highly autocorrelated samples of all four methods, and the lowest predictive performance.

9.4.1 Metropolis-Hastings and the Metropolis-Adjusted Langevin Algorithm II

MALA: adds first-order gradient information

MALA improves on MH by using the drift term $\frac{1}{2}f'(\mathbf{w})$ (Eq. 9.3) to bias proposals *uphill* in log-posterior, then Metropolis-corrects for the discretization error of the Euler–Maruyama integration. This directs exploration using local curvature (first-order gradient) rather than undirected random-walk steps, so MALA converges faster and mixes better than MH – though its samples remain correlated relative to Hamiltonian-based methods.

9.4.1 Metropolis-Hastings and the Metropolis-Adjusted Langevin Algorithm III

A note on manifold extensions

Riemannian-manifold variants of Langevin and Hamiltonian dynamics (Girolami & Calderhead, 2011) go further and use *second*-order gradient (curvature) information, at higher computational cost. In practice, methods using Hamiltonian dynamics on the ordinary Euclidean manifold (as used in this chapter) have shown great promise and are considerably simpler to implement.

Python: From-Scratch MALA on the Toy Posterior

```
1 def grad_log_posterior(w, X, y):
2     z = X @ w
3     p = sigmoid(z)
4     return X.T @ (y - p) - w / TAU**2    # analytic gradient (fixed-alpha case)
5
6 def run_mala(n_iter, step, w0, X, y, seed=1):
7     rng = np.random.default_rng(seed)
8     w = w0.copy()
9     lp, grad = log_posterior(w, X, y), grad_log_posterior(w, X, y)
10    chain, n_accept = np.zeros((n_iter, len(w))), 0
11    for m in range(n_iter):
12        mean_fwd = w + 0.5 * step * grad    # Eq. 9.3 drift
13        prop = mean_fwd + np.sqrt(step) * rng.standard_normal(len(w))
14        lp_prop = log_posterior(prop, X, y)
15        grad_prop = grad_log_posterior(prop, X, y)
16        mean_bwd = prop + 0.5 * step * grad_prop
17        log_q_fwd = -np.sum((prop - mean_fwd) ** 2) / (2 * step)
18        log_q_bwd = -np.sum((w - mean_bwd) ** 2) / (2 * step)
19        log_alpha = (lp_prop + log_q_bwd) - (lp + log_q_fwd)    # MH correction
20        if np.log(rng.uniform()) < log_alpha:
21            w, lp, grad = prop, lp_prop, grad_prop
22            n_accept += 1
23        chain[m] = w
24    return chain, n_accept / n_iter
```

9.4.2 Hamiltonian Monte Carlo – Efficient Recap I

Leapfrog integration (Eq. 9.5)

$$\mathbf{p}_{t+\epsilon/2} = \mathbf{p}_t + \frac{\epsilon}{2} \frac{\partial H(\mathbf{w}_t, \mathbf{p}_t)}{\partial \mathbf{w}}$$

$$\mathbf{w}_{t+\epsilon} = \mathbf{w}_t + \epsilon M^{-1} \mathbf{p}_{t+\epsilon/2}$$

$$\mathbf{p}_{t+\epsilon} = \mathbf{p}_{t+\epsilon/2} + \frac{\epsilon}{2} \frac{\partial H(\mathbf{w}_{t+\epsilon}, \mathbf{p}_{t+\epsilon/2})}{\partial \mathbf{w}}$$

where ϵ is the discretization step size and M is the covariance matrix of the (Gaussian) kinetic energy term. This is a symmetric half-step/full-step/half-step split that approximately conserves H .

9.4.2 Hamiltonian Monte Carlo – Efficient Recap II

Metropolis correction (Eq. 9.6)

$$\mathbb{P}(\text{accept } \mathbf{w}^*) = \min\left(1, \frac{\exp(-H(\mathbf{w}^*, \mathbf{p}^*))}{\exp(-H(\mathbf{w}, \mathbf{p}))}\right)$$

Needed because the leapfrog integrator is not exact – this restores exact detailed balance. HMC's two tunable parameters are the step size ϵ (usually tuned via dual averaging to hit a target acceptance rate) and the trajectory length (number of leapfrog steps) – a harder parameter to tune, often set via schemes such as the No-U-Turn Sampler.

9.4.2 Hamiltonian Monte Carlo – Efficient Recap III

Why HMC helps here

By simulating energy-conserving Hamiltonian trajectories rather than undirected or purely-gradient-biased single steps, HMC can propose **distant** points in parameter space that are still likely to be accepted. This is what drives its low autocorrelation – and, per this chapter's results, its best-in-class predictive performance on the real municipal dataset.

9.4.2 Magnetic Hamiltonian Monte Carlo (MHMC) I

Intuition, restated precisely. MHMC adds a **magnetic field** matrix G to the force field already present in HMC. This makes MHMC more flexible than HMC and promotes more efficient investigation of the posterior by reducing autocorrelation in the produced samples – *if* the magnetic field interacts favorably with the force field (e.g. if they are close to orthogonal).

Non-canonical Hamiltonian dynamics (Eq. 9.7)

$$\frac{d}{dt} \begin{bmatrix} \mathbf{w} \\ \mathbf{p} \end{bmatrix} = \begin{bmatrix} 0 & I \\ -I & G \end{bmatrix} \begin{bmatrix} \nabla_{\mathbf{w}} H(\mathbf{w}, \mathbf{p}) \\ \nabla_{\mathbf{p}} H(\mathbf{w}, \mathbf{p}) \end{bmatrix}$$

where G is a **skew-symmetric** (antisymmetric) matrix, i.e. $G^T = -G$, and is the term that represents the magnetic field. The Hamiltonian H itself is **identical** to the one used in ordinary HMC – only the matrix multiplying the gradient changes.

9.4.2 Magnetic Hamiltonian Monte Carlo (MHMC) II

Step-by-step: exactly what changes vs. ordinary HMC

Writing out both block rows of Eq. 9.7 and comparing to Eq. 9.4:

- **Position update (unchanged):** $\frac{d\mathbf{w}}{dt} = \nabla_{\mathbf{p}}H(\mathbf{w}, \mathbf{p})$ – identical in HMC and MHMC.
- **Momentum update (this is where the new term enters):**

$$\text{HMC: } \frac{d\mathbf{p}}{dt} = -\nabla_{\mathbf{w}}H(\mathbf{w}, \mathbf{p})$$

$$\text{MHMC: } \frac{d\mathbf{p}}{dt} = -\nabla_{\mathbf{w}}H(\mathbf{w}, \mathbf{p}) + G \nabla_{\mathbf{p}}H(\mathbf{w}, \mathbf{p})$$

The added term $G \nabla_{\mathbf{p}}H(\mathbf{w}, \mathbf{p})$ is exactly the extra “sideways” force: since G is skew-symmetric, $\mathbf{p}^\top (G\mathbf{p}) = 0$, so this term can never add or remove energy (it is orthogonal to $\mathbf{p}$ itself) – it can only *redirect* the momentum. **HMC and MHMC dynamics are identical whenever $G = 0$.**

9.4.2 Magnetic Hamiltonian Monte Carlo (MHMC) III

Setting the magnetic field G (as done in this chapter)

There is no general theory for how to choose G ; the authors follow Tripuraneni et al. (2017) and let only a *few* dimensions interact with the field. Concretely, for weight indices 1 and 5: $G_{1i} = G_{5i} = g$ and $G_{i1} = G_{i5} = -g$ for $g = 0.2$, zero elsewhere – and G is kept anti-symmetric throughout. Like HMC, the resulting dynamics cannot be integrated exactly, so a numerical integrator plus a Metropolis-Hastings accept step is used to enforce detailed balance despite discretization error.

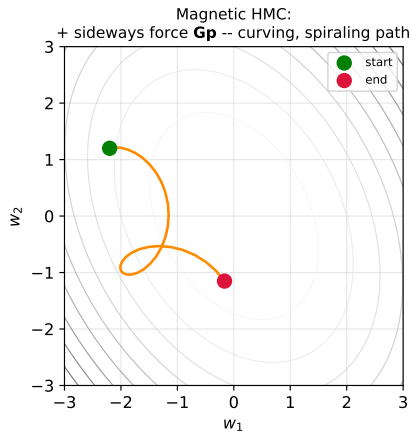
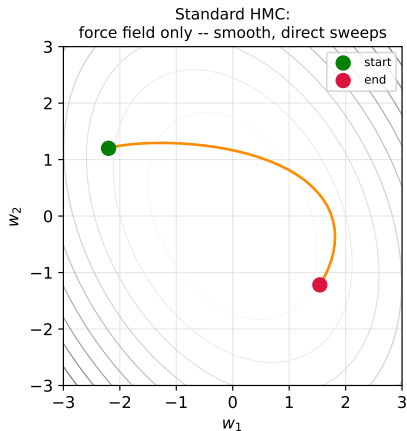
9.4.2 Magnetic Hamiltonian Monte Carlo (MHMC) IV

Physical analogy: Newtonian mechanics of a charged particle

According to Tripuraneni et al. (2017), these non-canonical dynamics represent the motion of a charged particle in a magnetic field in three dimensions. The particle still feels the ordinary potential force $(-\nabla_{\mathbf{w}}H)$, but a magnetic-type Lorentz force $(G \nabla_{\mathbf{p}}H)$ bends its path sideways without changing its speed. Whether MHMC actually samples *more* efficiently than HMC on a given posterior depends entirely on how this magnetic field geometrically interacts with that posterior's force field – there is no guarantee it always helps, and (as we will see in Sect. 9.6) it does not always help in practice.

Schematic: HMC vs. Magnetic HMC Trajectories (Toy 2D Potential)

Schematic: same toy potential, same starting momentum --
the magnetic term bends the trajectory sideways (illustrative)



Python: Leapfrog for HMC vs. Magnetic HMC (Toy Potential)

```
1 def grad_U(w, A):           # potential force: -grad(log posterior) = A @ w
2     return A @ w
3
4 def leapfrog_hmc(w0, p0, eps, L, A):
5     w, p = w0.copy(), p0.copy()
6     for _ in range(L):
7         p = p - 0.5 * eps * grad_U(w, A)      # HMC momentum half-step
8         w = w + eps * p                       # position full-step
9         p = p - 0.5 * eps * grad_U(w, A)      # HMC momentum half-step
10    return w, p
11
12 def leapfrog_mhmc(w0, p0, eps, L, A, G):
13     w, p = w0.copy(), p0.copy()
14     for _ in range(L):
15         force = -grad_U(w, A) + G @ p         # ADDED sideways term: G @ p
16         p = p + 0.5 * eps * force
17         w = w + eps * p
18         force = -grad_U(w, A) + G @ p         # recompute with new w, p
19         p = p + 0.5 * eps * force
20    return w, p
21
22 # G must be skew-symmetric: G.T == -G (e.g. G = [[0, g], [-g, 0]])
```

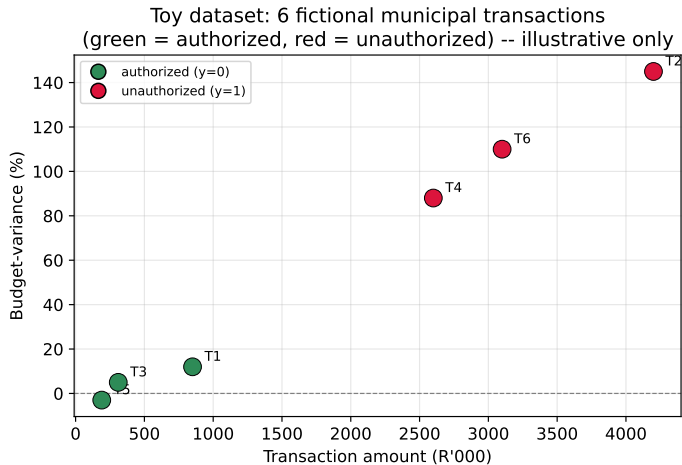
Toy Running Example: 6 Fictional Government Transactions I

Purpose. To make Sect. 9.3–9.4 concrete, we build our own small, fictional dataset in the same spirit as the book's real one, and fit the ARD Bayesian logistic regression from scratch. **These numbers are entirely illustrative** – they are not from the book's real municipal dataset.

ID	Description	Department	Amount (R)	Budget var. (%)	Label
T1	Roads resurfacing contract	Infrastructure	850,000	+12	authorized
T2	Emergency IT systems upgrade	IT Services	4,200,000	+145	unauthorized
T3	Community hall maintenance	Community Services	310,000	+5	authorized
T4	Bulk water pipeline repair	Water & Sanitation	2,600,000	+88	unauthorized
T5	Office stationery supply	Finance	190,000	−3	authorized
T6	Unbudgeted road-signage project	Roads & Transport	3,100,000	+110	unauthorized

Features used: standardized *amount* (x_1) and standardized *budget-variance* % (x_2) – i.e. how far actual spend deviated from the budgeted amount, our toy stand-in for the book's financial ratios.

Toy Running Example: 6 Fictional Government Transactions II



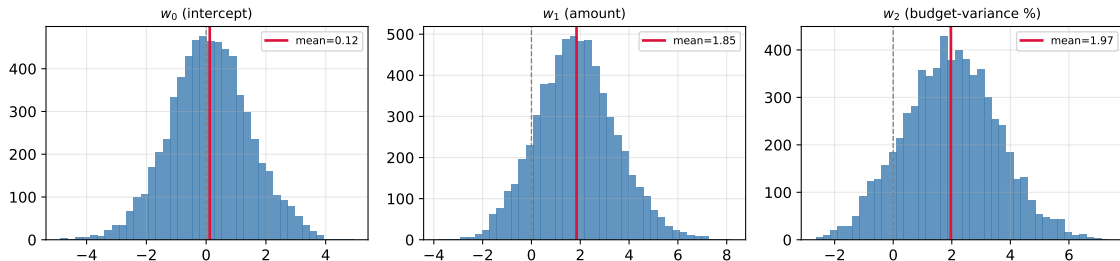
Toy Running Example: 6 Fictional Government Transactions III

Large, positive budget overruns coincide with unauthorized labels in this toy example – by design, since we made it up to be illustrative of the intuition, not to be a subtle classification problem.

Toy Running Example: Fitting the Model with MALA I

We fit the 3-parameter Bayesian logistic regression w_0 (intercept), w_1 (amount coefficient), w_2 (budget-variance-% coefficient) with $\mathcal{N}(0, \tau^2)$ priors ($\tau = 2$), using a from-scratch MALA sampler (8,000 iterations, burn-in 1,500, step size tuned to $\approx 92\%$ acceptance).

Toy example: posterior weights from from-scratch MALA
(3-parameter Bayesian logistic regression, illustrative only)



Toy Running Example: Fitting the Model with MALA II

Illustrative posterior summary (own toy simulation)

Parameter	Posterior mean	Posterior std. dev.
w_0 (intercept)	0.12	1.34
w_1 (amount)	1.85	1.62
w_2 (budget-variance %)	1.97	1.60

Both w_1 and w_2 have posterior mass concentrated well away from zero and of the same (positive) sign – consistent with our toy design that *larger amounts and larger positive budget overruns* both push a transaction towards “unauthorized.” With only 6 toy observations the posterior is necessarily wide (large std. devs.) – this is the Bayesian framework being honest about how little data it has seen, exactly the uncertainty quantification that a point-estimate classifier would hide.

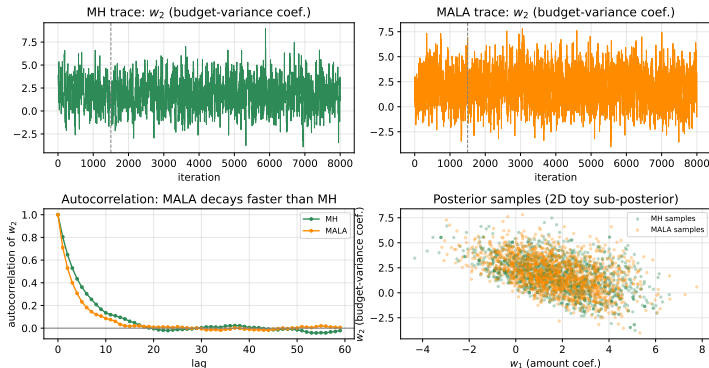
Toy Running Example: Fitting the Model with MALA III

A new toy transaction

For a new, unseen transaction of R2,000,000 with a +60% budget variance, plugging the posterior mean weights into the sigmoid link gives an illustrative $\mathbb{P}(\text{unauthorized}) \approx 0.57$ – i.e. the toy model is genuinely uncertain (just over a coin flip), which is a sensible answer for a transaction that sits between the clearly-authorized and clearly-unauthorized examples in our tiny training set.

Toy Running Example: MH vs. MALA on the Same Sub-Posterior

Toy example: from-scratch MH vs MALA on the same 2D
logistic-regression sub-posterior (illustrative only)



Own from-scratch simulation, illustrative only. MH tuned to $\approx 31\%$ acceptance, MALA to $\approx 83\%$. MALA's autocorrelation decays faster (bottom-left): rough ESS out of 6,500 post-burn-in draws is MH ≈ 663 vs. MALA ≈ 905 – a small-scale echo of the book's finding that gradient-informed samplers beat plain MH.

9.5.1 Data Description (the Book's Real Dataset) I

Source and scope

South African municipal finance dataset from the National Treasury website (municipaldata.treasury.gov.za). **1,430 observations** from 2012–2019, built from municipalities' *unaudited* financial statements as inputs, with the unauthorized-expenditure label taken from the corresponding *audited* financials.

Eight financial-ratio features

Debt to Community Wealth/Equity; Debt to Total Operating Revenue; Current Ratio; Net Operating Surplus Margin %; Remuneration % of Total Operating Expenditure; Contracted Services % of Total Operating Expenditure; Net Surplus/Deficit Water; Net Surplus/Deficit Electricity.

9.5.1 Data Description (the Book's Real Dataset) II

Class balance and preprocessing

When absolute expenditure amounts are converted to a binary label: 66% of observations have *no* unauthorized expenditure, 34% do. Data is split 70/30 into train/test. The distribution of absolute expenditure amounts is heavily right-skewed – most municipalities have (near) zero unauthorized expenditure, with a long tail of large violations, which is exactly why this is framed as a *binary detection* problem rather than a regression on the Rand amount.

9.5.2 Experiment Description

Task

Binary classification: does a given municipal financial report have unauthorized expenditure? Encoded as $y = 0$ (no) / $y = 1$ (yes).

Sampling setup

10,000 MCMC samples generated per method, of which the first 5,000 form the burn-in (used for step-size tuning). Step sizes tuned via primal-dual averaging (Hoffman & Gelman) to hit target acceptance rates of **25% (MH)**, **60% (MALA)**, **80% (HMC and MHMC)**. Trajectory length (number of leapfrog steps) for both HMC and MHMC fixed at **100**.

Python: Toy Experiment Harness (Same Structure as the Book)

```
1  # Illustrative harness mirroring the book's experimental protocol,  
2  # applied to our toy 6-transaction dataset (NOT the book's real data).  
3  N_SAMPLES, BURN_IN = 10_000, 5_000  
4  TARGET_ACCEPT = {"MH": 0.25, "MALA": 0.60, "HMC": 0.80, "MHMC": 0.80}  
5  TRAJECTORY_LENGTH = 100           # leapfrog steps, HMC and MHMC  
6  
7  def tune_step_size(sampler_fn, target_accept, X, y, **kwargs):  
8      """Simple bisection-style dual-averaging stand-in: adjust step until  
9      empirical acceptance rate is close to the target."""  
10     lo, hi = 1e-4, 5.0  
11     for _ in range(15):  
12         step = 0.5 * (lo + hi)  
13         _, acc = sampler_fn(n_iter=500, step=step, X=X, y=y, **kwargs)  
14         if acc > target_accept:  
15             lo = step           # accepted too often -> take bigger steps  
16         else:  
17             hi = step           # accepted too rarely -> take smaller steps  
18     return 0.5 * (lo + hi)
```

9.5.3 Performance Metrics for MCMC Methods

Diagnostics used in the book

- **Trace plots** of the un-normalized target log-posterior.
- **Effective sample size (ESS)** – the multivariate ESS measure of Vats et al. (2019), which accounts for autocorrelation across *all* sampled dimensions jointly.
- **ESS normalized by execution time** – ESS alone rewards cheap-but-slow-mixing methods unfairly; dividing by wall-clock time reveals which method delivers the most *useful* samples per second.
- **Predictive performance**: Receiver Operating Characteristic (ROC) curve and Area Under the ROC (AUC) on held-out test data – useful here because the relative cost of a false positive vs. a false negative audit flag is not clearly known in advance.

9.6 Results and Discussion I

Effective sample size: HMC wins outright

HMC produces the **highest** raw ESS, followed by MALA; MH and MHMC both have low, similar ESS. On a **time-normalized** basis, however, **MALA outperforms every method** – its per-iteration cost is far lower than the Hamiltonian-dynamics-driven methods, so despite mixing somewhat slower per-sample than HMC, it delivers more effective samples per second of compute. MH has the lowest execution time of all, followed by MALA then HMC; **MHMC has the largest execution time** due to the extra computation needed for the magnetic-field term.

Why MHMC underperformed here

All four methods converge (trace plots stabilize), but MHMC's poor ESS is attributed to a **poor choice of the magnetic field for this particular task** – i.e. the fixed, hand-set G (only coupling weights 1 and 5, with $g = 0.2$) likely does not interact favorably with this posterior's geometry. This points to the need to *tune* G rather than fix it a priori.

9.6 Results and Discussion II

Predictive performance (test-set AUC)

HMC: 0.7108 MALA: 0.7102 MHMC: 0.7085 MH: 0.6276

HMC has the best predictive performance (highest AUC), which aligns with its high effective sample size; MH has by far the poorest predictive performance, tracking its very poor exploration of the posterior (visible in its distinctly higher negative-log-likelihood trace and its inability to escape a worse mode).

Feature importance agreement

The three gradient-informed methods (MALA, HMC, MHMC) broadly **agree** on which financial ratios matter most, with **Current Ratio** and **Debt to Total Operating Revenue** the most consistently top-ranked features across methods – consistent with earlier audit-outcome analyses by the same authors. MH, lacking gradient information, cannot meaningfully distinguish important from unimportant features (its posterior variances for all 8 ratios come out similarly large).

9.7 Conclusions

Summary

- First-in-literature Bayesian ML framework for detecting unauthorized municipal expenditure from *pre-audit* (unaudited) financial ratios, using a Bayesian logistic regression with an ARD prior.
- Across MH, MALA, HMC, MHMC: **HMC** gives the best ESS and best predictive performance; **MALA** is the best performer once execution time is accounted for; **MH** is the clear laggard on every metric; **MHMC** needs better tuning of its magnetic field to realize its potential on this task.
- **Current ratio** and **debt to operating revenue** are consistently identified as the most important predictors of unauthorized expenditure.

Future work flagged by the authors

Extending the framework to predict the *absolute Rand value* of unauthorized expenditure (not just its presence/absence), and jointly modeling the other categories of unlawful spending – irregular, fruitless, and wasteful expenditure – alongside unauthorized expenditure.